

Tau3 Banking - Qwen 3.8 Max Trial-0 Parity Report

Agent-owned audit of the 97-task single-trial reproduction against the official Qwen 3.8 Max public trajectory artifact.

54/97 OFFICIAL PASSES	43/97 REPRODUCED PASSES	-11 PASS DELTA	97/97 VALID COVERAGE
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NOT EXACT

TL;DR

Official	Reproduced	Delta	Coverage	Main attention
54/97, 55.6701%	43/97, 44.3299%	-11 passes, -11.3402 percentage points	97/97, all <code>user_stop</code> , 0 infrastructure errors	Hosted sampling plus known OpenRouter user, embedding, and Modal transport differences

The reproduced aggregate does not match the official trial-0 score. The run is nevertheless a valid diagnostic: exact task coverage, configuration guards, model routes, protocol, deterministic regrading, and the task-102 judge route all passed, so the score gap is not a missing task or grader failure.

Parity evidence

Layer	Status	Evidence and nuance
Score	FAIL	43/97 vs 54/97. There are 23 task reward flips and 69 grading-component differences.
Harness	PARTIAL	Exact 97 tasks, trial 0, seed 626729, limits, agent arguments, and <code>alltools</code> retrieval were preserved. The official used local Linux <code>srt</code> ; the reproduction used the pinned Modal receipt , which is 4,603/4,614 unique commands exact and has 11 acknowledged model-visible residuals.
Tools	PARTIAL	Declared initial surface: 17 vs 17, schema <code>5e2d300ebc61227f1f79119768fe3c191a27896cbe844be7ccd49b0e9c271ae1</code> . Used: 20 vs 18 distinct names and 3,604 vs 3,840 calls. The candidate did not use <code>change_user_email</code> or <code>submit_transaction</code> . Strict comparison found 1,186 tool-call and 5,238 tool-output differences.
Model config	PARTIAL	Qwen request parity passed: <code>openrouter/qwen/qwen3.8-max</code> , Alibaba, xhigh reasoning, sole dated endpoint. The user simulator was necessarily OpenRouter-to-OpenAI GPT-5.2 instead of the official direct OpenAI transport.

Layer	Status	Evidence and nuance
Trace rendering	PARTIAL	Both traces use <code>tau2.to_litellm_messages</code> , OpenAI function-tool JSON, and <code>tool_choice=auto</code> . Qwen reasoning is retained in the current raw response and stripped from subsequent history. The provider tokenizer, Jinja chat template, BOS/EOS policy, and exact revision are not exposed, so they cannot be independently hashed.
Coverage	PASS	97/97 exact task and trial keys, 97 <code>user_stop</code> , 0 infrastructure errors, unique response IDs, and a stable clean runtime/cache fingerprint.
Grading	PASS	Offline DB, action, environment, communication, and reward recombination produced 0 integrity issues. Task 102 used the dated OpenAI GPT-4.1 judge and reproduced final reward 0, although its DB component differed.
Behavior	FAIL	6,424 strict behavior differences and 1,183 participant-message differences across all 97 simulations. The comparator attributes all 6,424 behavior differences to model sampling, but this cannot waive a nonmatching aggregate.

Trace boundary finding: 47 tasks had byte-identical first user text and the same first Qwen prompt-token count as official, yet 0/47 first Qwen outputs matched. Across all first GPT-5.2 calls, the OpenRouter trace was exactly 81 prompt tokens shorter on 81 tasks and equal on 16, confirming a transport boundary difference in addition to hosted nondeterminism. The Qwen route reported Alibaba for all 2,249 candidate generations; the GPT-5.2 user route reported OpenAI/default for all 652 generations.

Reward flips

Official pass to reproduced failure, 17 tasks:

`005,016,033,040,045,048,049,051,054,059,070,071,072,078,094,096,101`.

Official failure to reproduced pass, 6 tasks: `007,008,014,057,090,099`.

The net difference is -11 passes. Task 102 remained a final failure in both traces: official components were DB=1 and NL=0; reproduced components were DB=0 and NL=0. Its judge response was generated by `openrouter/openai/gpt-4.1-2025-04-14`, resolved to OpenAI with service tier `default`.

Secondary metrics derived from the public artifact

These use identical definitions on the official trial-0 slice and the complete candidate artifact. Chat cost excludes embeddings, Modal, failed-attempt spend, and the NL judge; the reproduced judge added \$0.10739.

Metric	Official	Reproduced	Delta
Generated participant messages	2,909	2,901	-8
Generated participant messages / simulation	29.9897	29.9072	-0.0825
Tool calls	3,604	3,840	+236
Tool calls / simulation	37.1546	39.5876	+2.4330
Distinct observed tool names	20	18	-2
Mean duration / simulation	393.5313 s	428.2069 s	+34.6756 s

Metric	Official	Reproduced	Delta
Participant chat cost	\$62.2505533	\$73.70333065	+\$11.45277735
Participant chat cost / simulation	\$0.6417583	\$0.7598282	+\$0.1180699
Qwen completion tokens / generation	677.3438	734.2939	+56.9501
Qwen reasoning tokens / generation	512.5318	578.6910	+66.1592

One live task-level restart occurred for task 054 after Qwen returned an empty final assistant message with neither content nor tool calls. The pinned upstream harness retried the whole simulation and the final checkpoint contains only the completed `user_stop` attempt; discarded-attempt spend is therefore not included in the serialized cost. This retry is disclosed as process non-parity because the lean parity contract permits only typed infrastructure retries.

Exact commands

All commands run from `/home/tianhaowu/mono/tau3-banking` on the Linux evaluation host. The credential is loaded from `~/.rllm/config.json` and never appears in argv, logs, manifests, or this report.

Launch the guarded 97-task x1 evaluation

```
ALLOW_FULL_RUN=1 uv run --frozen --extra knowledge python \
  reproduction/tau3_banking/run.py full \
  --output-dir reproduction/tau3_banking/runs/qwen38_full_trial0_repeat \
  --execute --confirm-paid-api-calls --cost-ceiling-usd 150 \
  --allow-known-full-shell-drift
```

- `full` means the frozen 97-task trial-0 slice, not the public 97 x 4 aggregate.
- `--confirm-paid-api-calls` acknowledges model, user-simulator, and judge calls.
- `--cost-ceiling-usd 150` is a historical-cost acknowledgement, not a provider hard cap.
- `--allow-known-full-shell-drift` acknowledges the 11 classified Modal shell residuals.
- The authenticated preflight required at least the official \$62.2505533 participant-chat cost before creating the run.
- Resume only the same immutable output manifest by appending `--resume`; never change config or copy scored outputs.

Run the strict offline comparison

```
uv run --offline --frozen --extra knowledge python \
  reproduction/tau3_banking/compare_results.py \
  reproduction/tau3_banking/runs/qwen38_full_trial0_repeat/results.json \
  --mode full \
  --output reproduction/tau3_banking/runs/qwen38_full_trial0_repeat/parity.json
```

Exit 1 is expected here because score, task vector, components, tools, and text do not match. Configuration, route, manifest, coverage, protocol, and grading-integrity fields must still pass.

Regenerate the compact metrics receipt

```
python3 reproduction/tau3_banking/analyze_qwen38_trial0.py \
  --reference reproduction/tau3_banking/artifacts/banking_knowledge_results.json \
  --candidate reproduction/tau3_banking/runs/qwen38_full_trial0_repeat/results.json \
  --parity reproduction/tau3_banking/runs/qwen38_full_trial0_repeat/parity.json \
  --output reproduction/tau3_banking/reports/qwen38_trial0_metrics.json
```

Render and verify this linked PDF

```
uv run reproduction/tau3_banking/render_parity_report.py \
  reproduction/tau3_banking/reports/qwen38_trial0_parity.md \
  reproduction/tau3_banking/reports/qwen38_trial0_parity.pdf \
  --pages-dir reproduction/tau3_banking/reports/qwen38_trial0_parity-pages
```

This one command creates the PDF, verifies all external URI annotations and extractable text, and renders every page to PNG for visual inspection.

Links and attention

- [Benchmark repository at the official commit](#), [leaderboard](#), and [raw official trajectory artifact](#)
- [Qwen 3.8 Max](#), [current endpoints](#), and [official submission PR](#)
- [Official tasks](#), [tool implementation](#), and [grader implementation](#)
- [Frozen benchmark config](#), [run manifest](#), and [metrics receipt](#)
- [Reproduction log](#), [standalone instructions](#), and [PDF QA receipt](#)

Attention: The available OpenRouter-only credential cannot reproduce the official direct-OpenAI GPT-5.2 user transport or the unrecoverable official dense-embedding cache. OpenRouter also does not expose the provider tokenizer revision or Jinja chat template. A coincidental future 54/97 would establish aggregate score parity only; it would not erase these trace and transport differences.

Hashes: official result `8c8191c43dfb2d21c1322cc154740e5e6044151837ab817c2cbbcd13ffeb626e`; reproduced result `92051d8923946012e609ad1f25cbc6f57754f8a49746e28fc8b9babb3a93e2d3`; strict comparator `fce8b34d20aff5745315910623185d453aae68837bd9f7c0e16a6c50e8d9accb`; run manifest `ea174dcb8464014752ec3f0cbc88415eb0a3de2d1ace7bc6267c565deb2731ce`.

Evaluation runtime commit `e28b25b8719e743a7ec7f36bb6dbf3a0b5d59bad`; evaluation ran from 2026-08-23 00:58:35 UTC to 02:16:10 UTC.